MILITARY SPECIFICATION SHEET

ELECTRON TUBES, RECEIVING

TYPES 1S5 AND 1U5

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Diode - Pentode

Outline --- 5-2 (EIA)

Base --- E7-1

Envelope --- T5-1/2

Cathode --- Coated filament

Base connections:

Pin No.	1	2	3	4	5	6	7
Element							
Type 1S5 ---	g3,-f	nc	1a (diode)	g2	2a (pentode)	g1	+f
Type 1U5 ---	-f,g3	2a (pentode)	g2	1a (diode)	nc	g1	+f

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec2	Ec1	Ilb(diode)	Ik
Unit:	Vdc	Vdc	Vdc	Vdc	μAdc	mAdc
Maximum:	1.6	100	100	0	250	5
Minimum:	1.2	---	---	---	---	---
TEST CONDITIONS:	1.4	67.5	67.5	0	---	---

GENERAL:

Qualification - Not required

Ⓓ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1266	Total grid current	1	Ec1 = -2.5 Vdc; Eb = Ec2 = 90 Vdc	0.65	II	Ic1	0	-0.5	µAdc
1256	Electrode current (anode)			0.65	II	Ib	1.05	2.15	mAdc
1306	Transconductance (2)	1	Ef = 1.1 Vdc	0.65	II	Sm	380	---	µmhos
1321	AC amplification		Ebb = Ecc2 = 45 Vdc; Rg2 = 2.0 Meg; Rg1 = 10 Meg; Rp = 0.5 Meg; Esig = 0.2 Vac; 0.1 µF between g2 and -f	0.65	II	Ep	6.5	---	Vac
1231	Emission (diode)		Eb = 10 Vdc	0.65	II	Ils	0.5	---	mAdc
Ⓓ 1246	Audio frequency noise	2	Ecal (0 Va)= 600 mVac; Ecal (50 mW) = 200 mVac	1.0	II	EB	---	---	vu
Ⓟ 1201	Short and discontinuity detection		40 G	0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration		Rp = 10,000 ohms	---	---	Ep	---	400	mVac
1301	Filament current			---	---	If	44	56	mAdc
1256	Electrode current (screen)			---	---	Ic2	0.24	0.56	mAdc
1306	Transconductance (1)			---	---	Sm	480	700	µmhos
1211	Insulation of electrodes			4.0	S3	---	---	---	---
1105	Permanence of marking			---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test		Group A; Ef = 1.4 Vdc or Vac (with equivalent Ec1); Ib = 250 µAdc	---	---	---	---	---	---
---	Life-test end points (500 hours):								
1306	Transconductance (1)or			---	---	Sm	420	---	µmhos
1321	AC amplification and			---	---	Ep	5.0	---	Vac
1231	Emission (diode)			---	---	Ils	0.2	---	mAdc

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NOTES:

1. This test to be performed at the conclusion of the holding period.

2. Ebb = Ecc2 = 135 Vdc; Rp = 1.0 Meg; Ecl = 0; Rgl = 1.5 Meg; Rg2 = 4.7 Meg (by-passed with 0.1 µF capacitor to -f). Rp of diode = 2 Meg to ground. Couple anode of diode to gl through a 0.1 µF capacitor.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:
Army - AR
Navy - AS, CG, OS, MC

Preparing activity:
Army - EL

Agent:
DLA - ES

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